

Popularising Summary

Louis Vandenbruwaene

Master's thesis: an extremal problem on connectivity in networks (Erdős Problem 915).

Networks are useful because they offer backup routes. If one road is closed, traffic takes another. If one cable fails, a message still arrives along a different chain of links. Mathematicians study this with graphs, where dots stand for objects and lines for the connections between them. This thesis asks a simple-sounding question: how many connections can a network with a fixed number of points have, if no two of those points may be joined by too many separate, non-overlapping routes?

This problem was posed by Béla Bollobás and Paul Erdős, listed today as Erdős Problem 915. In its original form it concerns ordinary networks and routes that may not share a connection, and for that case the answer has been known since a theorem of Mader. This thesis starts from that result and then changes the rules of the network one at a time: connections may be doubled up, directions may be added so that a link works only one way, or a single connection may join more than two points at once. The variants need different proofs and computational checks.

At that point the thesis hands the work to a computer. One program describes every version of the network in the same way, measures exactly how many separate routes run between any two points, and proves the answer outright for small networks by checking every case. For larger networks, it searches for good examples. Ordinary links use a short memory of recent moves to avoid immediate reversals, sometimes accepting worse states to escape a dead end. Many-point connections are added in shuffled order whenever feasible, restarting to explore other choices. When the search is pointed at cases we already understand, it rediscovers the very networks the earlier proofs singled out.

The thesis combines these experiments with mathematical proofs. For networks in which each connection joins several points, it determines the maximum under the strictest route limit and proves an upper bound under the next limit, attained in specified ranges. For one-way networks with ordinary links, it gives explicit upper bounds and dense constructions. The exact answer at larger route limits remains open, and a counterexample rules out an earlier formula.

For one-way networks whose links may be repeated, the thesis proves the exact maximum at every size and route limit. The proof finds a thin skeleton that preserves who can reach whom. Removing one such skeleton lowers the route limit, and repeating this process bounds the total number of links. For one-way networks with many-point connections, checked arguments determine the leading growth rate, while the exact finite values remain open.

AI tools assisted with code, writing and proof exploration. Arguments the author has not checked line by line are explicitly labelled as conjectures with proposed proofs. They concern, among other questions, whether larger networks can be optimised by assembling the best small pieces and how to classify all networks attaining an extremal value. These arguments are kept separate from the checked theorems. The searches supply examples and finite evidence, without claiming that a successful search proves a general formula.